In many cases, this limit exists and have a finite value. Finding the solution of such limits are called evaluation of the indeterminate form.

The indeterminate forms are $\frac{0}{0}, \frac{\infty}{\infty}$, $\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, \infty - \infty, 0^0, \infty^0$ and 1^∞.

Among all these indeterminate forms, $\frac{0}{0}$ is the fundamental one.

5.2.6 Methods of evaluation of algebraic limits

(i) Direct substitution

(ii) Factorization

(iii) Rationalisation

(iv) By using some standard limits

5.2.7 Some standard limits

(i) $\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}$ if $n \in Q$

(ii) $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = \lim_{\theta \to 0} \frac{\tan\theta}{\theta} = 1$, θ is in radian.

(iii) $\lim_{x \to 0} \frac{a^x - 1}{x} = \log_e a$ if $a > 0$

(iv) $\lim_{x \to 0} \frac{\log(1+x)}{x} = 1$

(v) $\lim_{x \to 0} (1+x)^{\frac{1}{x}} = e$

(vi) $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e$

(vii) $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$

Example 5.16

Evaluate: $\lim_{x \to 1} (3x^2 + 4x - 5)$.

Solution

$\lim_{x \to 1} (3x^2 + 4x - 5)$

$= 3(1)^2 + 4(1) - 5 = 2$

Example 5.17

Evaluate: $\lim_{x \to 2} \frac{x^2 - 4x + 6}{x + 2}$

Solution

$$\lim_{x \to 2} \frac{x^2 - 4x + 6}{x + 2} = \frac{\lim_{x \to 2}(x^2 - 4x + 6)}{\lim_{x \to 2}(x + 2)}$$

$$= \frac{(2)^2 - 4(2) + 6}{2 + 2} = \frac{1}{2}$$

Example 5.18

Evaluate : $\lim_{x \to \frac{\pi}{4}} \frac{5\sin 2x - 2\cos 2x}{3\cos 2x + 2\sin 2x}$

Solution

$$\lim_{x \to \frac{\pi}{4}} \frac{5\sin 2x - 2\cos 2x}{3\cos 2x + 2\sin 2x}$$

$$= \frac{\lim_{x \to \frac{\pi}{4}} (5\sin 2x - 2\cos 2x)}{\lim_{x \to \frac{\pi}{4}} (3\cos 2x + 2\sin 2x)}$$

$$= \frac{5\sin\frac{\pi}{2} - 2\cos\frac{\pi}{2}}{3\cos\frac{\pi}{2} + 2\sin\frac{\pi}{2}}$$

$$= \frac{5(1) - 2(0)}{3(0) + 2(1)} = \frac{5}{2}$$

Example 5.19

Evaluate: $\lim_{x \to 1} \frac{x^3 - 1}{x - 1}$

Solution

$\lim_{x \to 1} \frac{x^3 - 1}{x - 1}$ is of the type $\frac{0}{0}$

$$\lim_{x \to 1} \frac{x^3 - 1}{x - 1} = \lim_{x \to 1} \frac{(x-1)(x^2 + x + 1)}{(x - 1)}$$

$$= \lim_{x \to 1} (x^2 + x + 1)$$

$$= (1)^2 + 1 + 1$$

$$= 3$$

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